

Bio Informatics

Lecture 17

Multiple Sequence Alignment – Progressive Alignments

- Progress alignment is first proposed by Feng and Doolittle (1987).
- It is a heuristics to get a good multiple alignment.
- Basic idea:
 - Align the two most closest sequences
 - Progressive align the most closest related sequences until all sequences are aligned.
- Examples of Progress alignment method include:
 - ClustalW, T-coffee, Probcons
- Probcons is currently the most accurate MSA algorithm.
- ClustalW is the most popular software.

Progressive Alignments – Basic Algorithm

1. Computing pairwise distance scores for all pairs of sequences
2. Generate the guide tree which ensures similar sequences are nearer in the tree
3. Aligning the sequences one by one according to the guide tree

Progressive Alignment – Example

- S1 : PPGVKSDCAS
- S2 : PADGVKDCAS
- S3 : PPDGKSDS
- S4 : GADGKDCCS
- S5 : GADGKDCAS

Pairwise Distance Scores (Step-1)

- Example: For S_1 and S_2 , the global alignment is
 - $S_1 = \text{P-PGVKSDCAS}$
 - $S_2 = \text{PADGVK-DCAS}$
- There are 9 non-gap positions and 8 match positions.
- The distance is $1 - 8/9 = 0.111$

Pairwise Distance Scores (Step-1)

	S_1	S_2	S_3	S_4	S_5
S_1	0	0.111			
S_2		0			
S_3			0		
S_4				0	
S_5					0

$S_1 = \text{P-PGVKSDCAS}$

$S_2 = \text{PADGVK-DCAS}$

S_1 : PPGVKSDCAS

S_2 : PADGVKDCAS

S_3 : PPDGKSDS

S_4 : GADGKDCCS

S_5 : GADGKDCAS

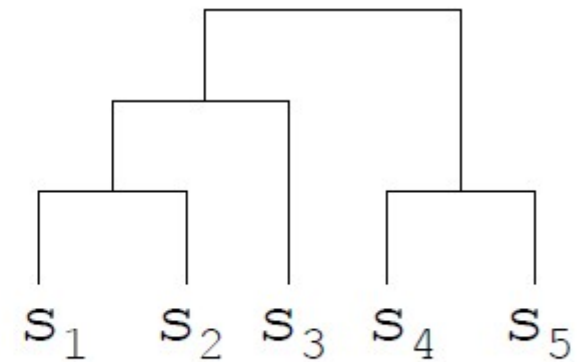
Pairwise Distance Scores (Step-1)

	S_1	S_2	S_3	S_4	S_5
S_1	0	0.111	0.25	0.555	0.444
S_2		0	0.375	0.222	0.111
S_3			0	0.5	0.5
S_4				0	0.111
S_5					0

Generate Guide Tree (Step-2)

- By neighbor-joining, generate the guide tree.

	S_1	S_2	S_3	S_4	S_5
S_1	0	0.111	0.25	0.555	0.444
S_2		0	0.375	0.222	0.111
S_3			0	0.5	0.5
S_4				0	0.111
S_5					0



Align sequences according to Guide Tree (Step-3)

- Aligning S_1 and S_2 , we get

- $S_1 = \text{P-PGVKSDCAS}$

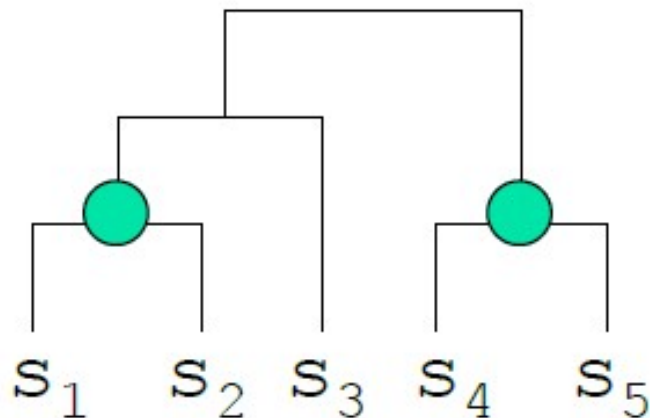
- $S_2 = \text{PADGVK-DCAS}$

- Aligning S_4 and S_5 ,

- $S_4 = \text{GADGKDCCS}$

- $S_5 = \text{GADGKDCAS}$

	S_1	S_2	S_3	S_4	S_5
S_1	0	0.111	0.25	0.555	0.444
S_2		0	0.375	0.222	0.111
S_3			0	0.5	0.5
S_4				0	0.111
S_5					0



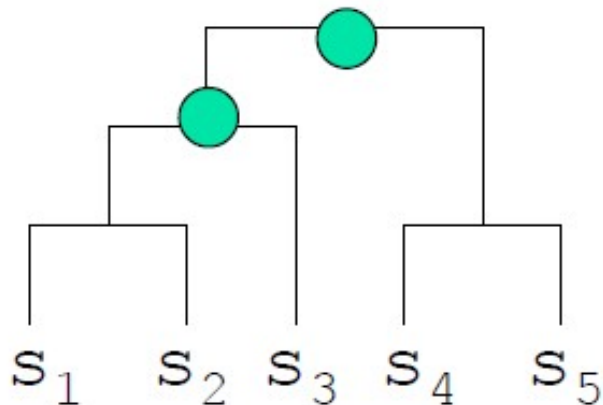
Align wrt Guide Tree(Step-3)

- Aligning (S1, S2) with S3, we get

- $S_1 = \text{P-PGVKSDCAS}$
- $S_2 = \text{PADGVK-DCAS}$
- $S_3 = \text{PPDG-KSD--S}$

- Aligning (S1, S2, S3) with (S4, S5), we get

- $S_1 = \text{P-PGVKSDCAS}$
- $S_2 = \text{PADGVK-DCAS}$
- $S_3 = \text{PPDG-KSD--S}$
- $S_4 = \text{GADG-K-DCCS}$
- $S_5 = \text{GADG-K-DCAS}$



S_1 : P-PGVKSDCAS
 S_2 : PADGVK-DCAS
 S_3 : PPDG-KSD--S
 S_4 : GADG-K-DCCS
 S_5 : GADG-K-DCAS

Summary

S_1 : PPGVKSDCAS
 S_2 : PADGVKDCAS
 S_3 : PPDGKSDS
 S_4 : GADGKDCCS
 S_5 : GADGKDCAS



	S_1	S_2	S_3	S_4	S_5
S_1	0	0.111	0.25	0.555	0.444
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S_3			0	0.5	0.5
S_4				0	0.111
S_5					0



S_1 : P-PGVKSDCAS
 S_2 : PADGVK-DCAS
 S_3 : PPDG-KSD--S
 S_4 : GADG-K-DCCS
 S_5 : GADG-K-DCAS

